APPLIED AI RESEARCH LABORATORY

We are an applied AI research lab focused on state-space models, RWKV, and scientific analysis use cases.

THESIS

What we are building

A company that monetizes quickly through AI-enabled services and compounds opted-in data into a long-term model advantage for spatial and scientific AI.

- State-space models and RWKV are the architectural entry point.
- We start at the data-format and representation layer.
- The long-term goal is a general foundation model for scientific use cases.

We combine architecture research with practical system-building. Our operating model is roughly 70% building and 30% research, with near-term monetization through AI-enabled services agreements.

State-space models. RWKV. Scientific analysis.

Services first. Representation layer next. Foundation model over time.

THE CORE PROBLEM

Science breaks generic AI because the world is not stored in one clean modality.

The hard problem is not only reasoning. It is the many different data formats, dimensions, coordinate systems, and physical assumptions required to communicate multi-spatial concepts across scientific workflows.

01

Heterogeneous formats

Scientific work is spread across incompatible file types, array layouts, tables, grids, coordinate frames, and discipline-specific conventions.

02

Hidden semantics

Spatial, temporal, and physical meaning often lives outside the raw tensor. General AI systems miss the semantics that scientists actually use.

03

Cross-disciplinary failure

Most current models operate within a single modality or domain. Scientific intelligence requires movement across representations and disciplines.

HOW WE MONETIZE

We plan to monetize fast through AI-enabled services agreements with a forward deployed engineering model.

The near-term business looks more like Palantir than a pure API company: embed deeply, solve hard technical workflows, and use that work to produce durable data, product insight, and domain trust.

OPERATING MODEL

70 / 30 mix

About 70% of effort goes to delivery and 30% to research. Revenue stays close to the work while technical edge keeps compounding.

NEAR TERM

FDE services

Start with forward deployed AI work inside real scientific workflows. Contracts fund execution, trust, and reusable implementation patterns.

LONG TERM

Compounding data

Each opted-in deployment sharpens product direction, representation requirements, and the long-run training corpus.

Budget planning

- Target 2-3 year service contracts at about $200k total contract value per customer.
- That implies roughly $67k-$100k in annualized revenue per account depending on term length.
- Keep research spend tied to services margin instead of a separate burn-heavy team.

Financial GTM expectations

- Land with forward deployed work, then expand into multi-year workflow ownership.
- 4-6 active contracts implies about $800k-$1.2M in contracted backlog and roughly $267k-$600k in annualized revenue.
- Prioritize scientific teams where deployments produce repeatable workflows and opt-in learning value.

STARTING POINT

The plan is to start at the data format layer.

Science's hard problem is that many different formats and dimensions are required to communicate multi-spatial concepts. If we cannot unify the representation layer, we cannot build truly general scientific AI.

01

Ingest

Capture heterogeneous scientific assets across formats, structures, and dimensions.

02

Normalize

Represent coordinates, arrays, tables, transforms, and metadata in a common computational form.

03

Learn

Train jointly over data structures, semantics, and domain transformations.

04

Generalize

Move toward models that operate across scientific disciplines instead of a single modality.

RESEARCH THESIS

We believe the missing layer for general scientific AI is a representation layer that unifies heterogeneous scientific data formats and their underlying spatial, temporal, and physical semantics.

01

What must be learned

The model should learn not only values, but also data structures, coordinate systems, domain transformations, and the relations between them.

02

What becomes possible

By learning jointly over structure and semantics, a model can reason across disciplines rather than being trapped inside one modality or file type.

03

Why this matters

A general scientific foundation model will need to move fluidly through spatial, temporal, and physical representations that today's models mostly ignore.

This is the bridge from applied services to defensible model IP: every deployment teaches us more about the representations required for cross-disciplinary scientific intelligence.

VISION

~~Our~~ goal is to build the first general foundation model for spatial and scientific use cases.

We will get there by mixing a build-heavy operating model with focused research, monetizing quickly through forward deployed AI services, and compounding toward a unified representation layer for scientific intelligence.

REVENUE

Near-term company

AI-enabled services agreements, forward deployed engineering, and technical workflow ownership.

MOAT

Compounding asset

Opted-in data and domain knowledge that improve the representation layer and future model training.

VISION

Long-term outcome

A foundation model capable of operating across scientific disciplines rather than within a single modality.

Beag Labs. State-space models. RWKV. Scientific analysis.

https://www.beaglabs.com/

https://github.com/beaglabs